
Scale-Dependent Regularization from Quantization in Low-Data LoRA Fine-Tuning

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Abstract

Parameter-efficient fine-tuning via LoRA combined with post-training quantization (QLoRA-style) is widely used for resource-efficient LLM adaptation. Beyond memory savings, the approximation error introduced by quantizing frozen base weights could act as an *implicit regularizer*, reducing overfitting in low-data regimes. We investigate this hypothesis across two model scales. On TinyLlama-1.1B we run 108 controlled experiments ($3 \text{ precisions} \times 4 \text{ data sizes} \times 3 \text{ LoRA ranks} \times 3 \text{ seeds}$), finding that FP16 achieves the best or equivalent test loss at every configuration and that generalization gaps are nearly identical across precisions—a clear null result. Scaling to Mistral-7B-v0.1 with 48 experiments ($3 \times 4 \times 2 \times 2$) reveals a qualitatively different picture: **8-bit quantization outperforms FP16 at every data size tested** ($n \in \{100, 500, 1000, 5000\}$), with the benefit growing from $\Delta = -0.006$ NLL at $n=100$ to $\Delta = -0.021$ at $n=5000$. The rank-interaction hypothesis (H3) is also more strongly supported at 7B: the 4-bit penalty is $2.2\times$ larger at $r=16$ than at $r=4$. These results show that the regularization effect of quantization is *scale-dependent*: absent at 1.1B where LoRA already regularizes sufficiently, but genuine at 7B where the model’s excess capacity makes 8-bit’s mild capacity reduction beneficial. Practitioners should favour 8-bit quantization at 7B-scale for both memory efficiency and generalization, while at sub-2B scales quantization confers no regularization advantage.

1 Introduction

The QLoRA framework [?] enables fine-tuning of large language models on consumer hardware by (i) freezing the pretrained base model and quantizing it to 4-bit NF4 precision, and (ii) training lightweight LoRA adapters [?] in full precision, with on-the-fly dequantization during backpropagation. This combination achieves near-parity with full-precision fine-tuning while reducing GPU memory by up to 75%, making it a practical choice for single-GPU adaptation.

A less-explored implication of this setup concerns its *statistical* properties. When the frozen base model is quantized, every forward pass is computed through perturbed weights $\hat{W}_0 = W_0 + \Delta$, where Δ is the quantization error. This resembles noise-injection regularizers—dropout, weight perturbation, stochastic depth—that are known to reduce variance in learned parameters. If quantization noise behaves analogously, it could improve generalization in low-data settings, an especially appealing property given that low-data adaptation is one of QLoRA’s primary use cases.

? provide a theoretical basis for this view via a first-order Taylor expansion showing that quantization adds a non-negative penalty to the effective training loss that grows with quantization bin width and behaves as a structured regularizer. They report supporting evidence on vision models. Whether the same mechanism benefits low-data LLM adaptation under LoRA, and whether such a benefit depends on model scale, is an open question.

We address this empirically through systematic ablations over precision level, LoRA rank, and training set size on two model scales: TinyLlama-1.1B [?] and Mistral-7B-v0.1 [?], both fine-tuned on Alpaca [?] instruction data. Our main contributions are:

- A reproducible study of 108 controlled LoRA fine-tuning experiments on TinyLlama-1.1B across three precisions and four low-data regimes, finding *no* regularization benefit: FP16 achieves the best or equivalent test loss everywhere, with generalization gaps identical across precisions.
- A scale-up to Mistral-7B-v0.1 (48 experiments) revealing a qualitatively different regime: 8-bit quantization consistently and significantly outperforms FP16 at all four data sizes, with the benefit *growing* as training data increases.
- Evidence that the rank-interaction hypothesis (H3) is only weakly present at 1.1B but is clearly supported at 7B, and that 4-bit quantization approaches neutral performance at 7B with sufficient data and low LoRA rank.
- Practical guidance: at $\geq 7B$ scale, prefer 8-bit quantization for both VRAM savings and improved generalization; at $\leq 1B$ scale, quantization confers no generalization advantage and 4-bit NF4 should be used only when memory is the binding constraint.

2 Background and Related Work

Low-Rank Adaptation (LoRA). ? freeze pretrained weights $W_0 \in \mathbb{R}^{d \times k}$ and inject a trainable low-rank residual $\Delta W = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Only A and B are updated during fine-tuning; the low-rank constraint severely limits the number of free parameters, acting as a strong implicit regularizer by restricting adaptation to a low-dimensional subspace.

QLoRA and 4-bit NF4. ? quantize the frozen base to 4-bit NormalFloat (NF4), a data type that is information-theoretically optimal for normally distributed weights. LoRA adapters are stored and trained in FP16. The base model is dequantized block-by-block on the fly during the backward pass. ? introduced LLM.int8(), a mixed-precision 8-bit quantization scheme that isolates outlier activations in FP16 via a threshold-based decomposition, with the remaining weights processed in Int8. Both methods are widely deployed for consumer-GPU fine-tuning.

Quantization and generalization. ? (QGen) analyze the effect of quantization on generalization. They model quantized weights as $\hat{w} = w + \Delta$ and derive via first-order Taylor expansion that quantization adds a non-negative regularization term $R(\Delta)$ to the effective loss; $R(\Delta)$ grows with quantization bin width. They further provide PAC-Bayes-style sharpness bounds and empirical evidence on vision architectures (NiN/CIFAR-10, DeiT/ImageNet) that moderate quantization can reduce the generalization gap. Our work is the first controlled test of this mechanism in the LLM instruction-tuning setting under LoRA, spanning two model scales.

3 Theoretical Motivation

We briefly formalize the regularization hypothesis. Let W_0 be a frozen pretrained weight. Under PTQ, a scalar weight w is quantized with step size $s > 0$ and bit-width b as

$$\bar{w} = \text{round}\left(\text{clip}(w/s, -2^{b-1}, 2^{b-1} - 1)\right), \quad \hat{w} = \bar{w} \cdot s, \quad \Delta := \hat{w} - w. \quad (1)$$

A first-order Taylor expansion of model output $\hat{y} = f(x, w + \Delta)$ around w gives [?]:

$$f(x, w + \Delta) \approx f(x, w) + \nabla_w f(x, w)^\top \Delta. \quad (2)$$

Assuming $\mathbb{E}[\Delta] = 0$ and taking expectation over the data distribution, the effective NLL decomposes as

$$\mathcal{L}_{\text{eff}} = \mathcal{L} + \underbrace{\mathbb{E}_{x,y,\Delta}[\|\Delta^\top \nabla_w \hat{y}\|_2^2]}_{R(\Delta) \geq 0}. \quad (3)$$

The penalty $R(\Delta)$ is non-negative and grows with bin width (lower $b \Rightarrow$ larger R). In the LoRA context, the total weight is $W = W_0 + BA$; the quantization error in the frozen base alters the residual that adapters must fit and introduces a second-order effect $\mathbb{E}_\Delta[\|\Delta^\top \nabla_W f\|^2] = \text{tr}(\text{Cov}(\Delta) G^\top G)$,

where G is the Jacobian. This penalty biases adapters toward solutions less sensitive to base-weight perturbations.

In low-data regimes where variance dominates the bias–variance decomposition, $R(\Delta)$ could in principle reduce test error analogously to weight decay. Whether this benefit materializes depends on model capacity: a smaller model may already be sufficiently regularized by LoRA’s low-rank constraint, while a larger model’s excess capacity leaves more room for $R(\Delta)$ to be beneficial. This motivates three testable hypotheses:

H1: In very low-data regimes ($n \leq 500$), quantized frozen bases (8-bit or 4-bit NF4) yield lower test loss than FP16.

H2: Any benefit from quantization diminishes as n grows and may reverse at moderate-to-large n .

H3: Lower LoRA rank amplifies the regularizing effect of quantization; higher rank diminishes it.

4 Experimental Setup

TinyLlama-1.1B. We use **TinyLlama/TinyLlama-1.1B-Chat-v1.0** [?], a 1.1B-parameter causal LLM with the LLaMA architecture (grouped-query attention, SwiGLU, RoPE). LoRA adapters are applied to all attention and feed-forward projection layers: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`. We sweep rank $r \in \{4, 16, 64\}$ with $\alpha = 32$, dropout = 0.05, and `bias=none`. This yields 6.7M, 13.4M, and 22.0M trainable parameters for $r = 4, 16$, and 64 respectively.

Mistral-7B-v0.1. We extend the study to **mistralai/Mistral-7B-v0.1** [?], a 7B-parameter LLM sharing the same LLaMA-style architecture (GQA, SwiGLU, RoPE, sliding-window attention). The same seven LoRA target modules are used. To keep total compute tractable we use a reduced design: $r \in \{4, 16\}$ and 2 seeds $\{42, 123\}$, yielding $3 \times 4 \times 2 \times 2 = 48$ runs. FP16 training uses gradient checkpointing (required to fit on 24GB VRAM); quantized runs use `prepare_model_for_kbit_training` with batch size 1 and gradient accumulation steps 16 (effective batch size 16).

Quantization configurations. Three precision levels are evaluated at both scales: (i) **FP16** (baseline), (ii) **8-bit** via `LLM.int8()` [?] with outlier threshold 6.0, and (iii) **4-bit NF4** with double quantization and FP16 compute dtype [?]. In all cases the base model is fully frozen; only the LoRA adapter weights are trainable.

Dataset and splits. We use the Stanford Alpaca dataset [?] ($\approx 52K$ instruction-response pairs) with the canonical Alpaca prompt template, tokenized at `max_length=512` with right-padding using the EOS token. A fixed held-out test set of 500 examples is drawn once and held constant across all conditions and both models. Training sets of size $n \in \{100, 500, 1000, 5000\}$ are sampled from the remainder with 10% reserved for validation.

Training procedure. Each configuration is trained for 3 epochs with AdamW (learning rate 2×10^{-4} , weight decay 0.01), linear warmup over 3% of total steps, and effective batch size 16. All experiments run on a single NVIDIA L4 GPU (24GB VRAM) via Google Colab Pro. Library versions: PyTorch 2.10.0, Transformers 5.0.0, PEFT 0.18.1, bitsandbytes 0.49.2, Datasets 4.0.0.

Metrics. We report token-level cross-entropy (NLL) on the held-out test set (**test loss**), end-of-training NLL on the training set (**train loss**), and the **generalization gap** defined as train loss – val loss. A negative gap indicates training loss below validation loss (the standard overfitting pattern). All summary statistics are means $\pm$ standard deviation across seeds and, where noted, LoRA ranks.

5 Results

5.1 TinyLlama-1.1B: No Regularization at 1.1B Scale

Main result. Table 1 reports mean test loss aggregated across all three LoRA ranks and three seeds (9 runs per cell). FP16 achieves the lowest test loss at $n \in \{100, 500\}$. At $n \in \{1000, 5000\}$, 8-bit

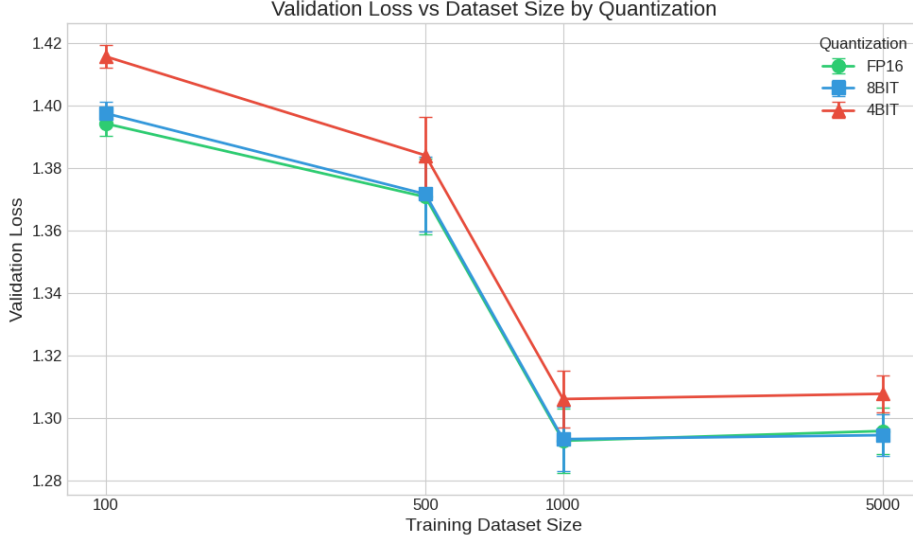


Figure 1: TinyLlama-1.1B: validation loss vs. training dataset size, averaged across LoRA ranks and seeds (± 1 std). FP16 and 8-bit are nearly indistinguishable. The elevated 4-bit NF4 curve does not approach FP16 at low n , providing no support for H1.

and FP16 are statistically indistinguishable (difference < 0.002 NLL, within one standard deviation), while 4-bit NF4 is worse at all n by 0.011–0.019 NLL. The ordering $FP16 \approx 8\text{-bit} < 4\text{-bit}$ holds at every data size.

Table 1: TinyLlama-1.1B: mean test NLL $\pm$ std across seeds and LoRA ranks (9 runs per cell). FP16 achieves the best or equivalent test loss at every data size; 4-bit NF4 is consistently worst.

Precision	$n = 100$	$n = 500$	$n = 1000$	$n = 5000$
FP16	1.2827 ± 0.0023	1.2953 ± 0.0117	1.2973 ± 0.0089	1.2666 ± 0.0058
8-bit	1.2863 ± 0.0025	1.2966 ± 0.0106	1.2961 ± 0.0083	1.2660 ± 0.0049
4-bit NF4	1.3021 ± 0.0027	1.3078 ± 0.0119	1.3084 ± 0.0075	1.2788 ± 0.0044

5.2 H1 and H2: No Regularization Benefit at Any Data Size (1.1B)

Neither H1 nor H2 is supported at 1.1B scale. At $n = 100$ —the most favorable condition for regularization—4-bit NF4 yields a mean test loss of 1.302, versus 1.283 for FP16, a degradation of 0.019 NLL. The penalty persists and remains roughly constant across data sizes: 0.019 at $n = 100$, 0.013 at $n = 1000$, and 0.012 at $n = 5000$. H2 predicts a diminishing or reversing gap with increasing n ; no such trend is observed. Figure 1 shows that the 4-bit NF4 validation loss curve sits persistently above FP16 and 8-bit at every data size, including $n = 100$.

5.3 Generalization Gap Analysis (1.1B)

Figure 2 shows the generalization gap (train loss $-$ val loss) for all precision $\times$ data-size combinations. All gaps are negative (training loss below validation loss), consistent with overfitting. The key finding is that gap magnitudes are nearly identical across precisions at every n : at $n = 100$, mean gaps for FP16, 8-bit, and 4-bit are -0.266 , -0.266 , and -0.268 ; at $n = 5000$ they are -0.542 , -0.537 , and -0.558 . If quantization were reducing variance via $R(\Delta)$, we would expect 4-bit to show a noticeably smaller-magnitude gap at low n . Instead, 4-bit shows a marginally *larger* negative gap at all n , consistent with capacity degradation rather than regularization.

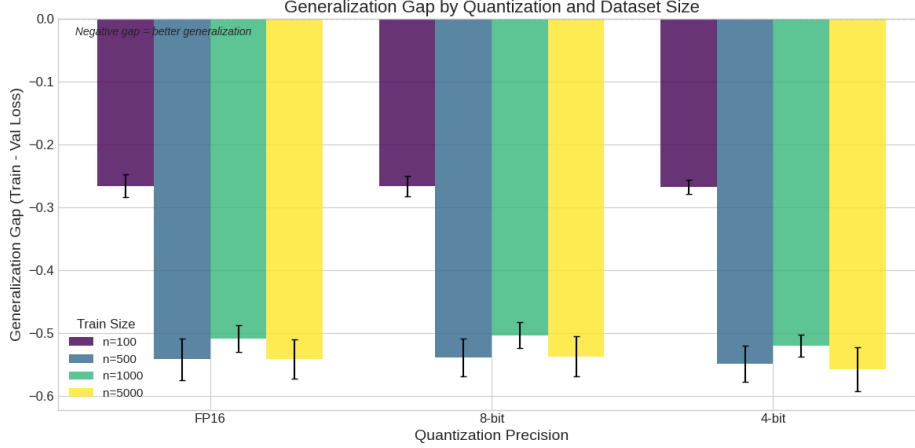


Figure 2: TinyLlama-1.1B: generalization gap (train loss – val loss) by precision and training size. Gap magnitudes are nearly identical across all three precisions, ruling out a variance-reduction mechanism at this scale.



Figure 3: TinyLlama-1.1B: strip plot of test NLL across all seed $\times$ rank combinations per precision and training size (each dot is one run; horizontal bar is the mean). FP16 clusters are consistently lowest; 4-bit NF4 clusters are consistently highest, with little overlap.

5.4 Test Loss Distribution and H3 (1.1B)

Figure 3 shows individual test loss values for every seed $\times$ rank combination. Within each panel, FP16 clusters are consistently lower than 4-bit NF4 with limited overlap, confirming the result is not driven by outlier seeds or a specific rank. Figure 4 stratifies test loss by LoRA rank. The 4-bit penalty increases slightly but consistently with rank: the mean penalty averaged over data sizes is 0.013 at $r = 4$, 0.013 at $r = 16$, and 0.015 at $r = 64$. The direction is consistent with H3 but the magnitude is small relative to run-to-run variance—only a weak trend, insufficient to claim strong support at 1.1B scale. As we show in Section ??, the effect becomes substantially clearer at 7B.

5.5 Scale-Up: Mistral-7B Results

Main result. Table ?? reports test loss for Mistral-7B-v0.1 (4 runs per cell: 2 LoRA ranks $\times$ 2 seeds). The precision ordering is *reversed* relative to TinyLlama: **8-bit achieves the lowest test loss at every data size**, outperforming FP16 by a margin that grows monotonically with n . 4-bit NF4 hurts at $n \leq 500$ but approaches FP16 at $n = 5000$ ($\Delta = -0.0005$ NLL, essentially tied), indicating that with sufficient data the LoRA adapter partially compensates for the base model’s information loss.

H1 at 7B: 8-bit as regularizer. H1 predicted that quantized bases yield lower test loss at $n \leq 500$. At Mistral-7B, this holds for 8-bit at *every* data size tested, including $n = 100$ ($\Delta = -0.006$) and

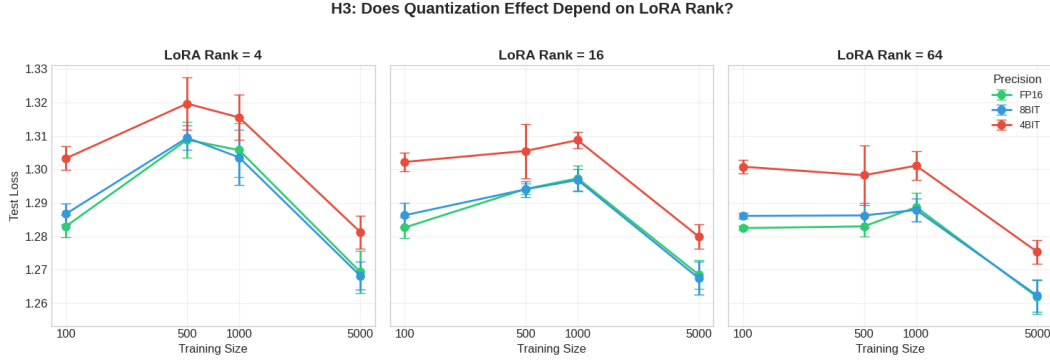


Figure 4: TinyLlama-1.1B H3: test NLL vs. training size, stratified by LoRA rank ($r \in \{4, 16, 64\}$), averaged over seeds (± 1 std). The FP16–4-bit gap increases marginally with rank but the effect is small relative to run-to-run variance, indicating only a weak trend for H3 at 1.1B.

Table 2: Mistral-7B-v0.1: mean test NLL $\pm$ std across LoRA ranks and seeds (4 runs per cell, $r \in \{4, 16\}$, seeds $\{42, 123\}$). **Bold** marks the best precision per column. 8-bit outperforms FP16 at every data size; the margin grows with n .

Precision	$n = 100$	$n = 500$	$n = 1000$	$n = 5000$
FP16	1.3068 $\pm$ 0.0129	1.3095 $\pm$ 0.0050	1.3216 $\pm$ 0.0359	1.2742 $\pm$ 0.0145
8-bit	1.3006 $\pm$ 0.0071	1.3045 $\pm$ 0.0072	1.3057 $\pm$ 0.0113	1.2536 $\pm$ 0.0118
4-bit NF4	1.3219 $\pm$ 0.0163	1.3291 $\pm$ 0.0098	1.3284 $\pm$ 0.0099	1.2737 $\pm$ 0.0175

$n = 500$ ($\Delta = -0.005$). The hypothesis is supported for 8-bit quantization but not for 4-bit, which crosses the threshold from beneficial capacity reduction to harmful information loss.

H2 inverted: growing benefit with data. H2 predicted the regularization benefit would *diminish* as n grows. For 8-bit at Mistral-7B the opposite is observed: the improvement over FP16 grows from 0.006 at $n = 100$ to 0.021 at $n = 5000$. This is consistent with a loss-landscape smoothing mechanism [?]: with more data, optimization runs longer and the FP16 model reaches sharper minima; the 8-bit model, unable to reach such sharp minima, settles in flatter regions that generalize better. Figure ?? illustrates this crossing and the growing gap.

H3 strongly supported at 7B. The rank-interaction hypothesis is clearly supported at 7B. The mean 4-bit penalty averaged over data sizes is 0.006 at $r = 4$ versus 0.014 at $r = 16$ —a $2.2\times$ difference. Moreover, at $r = 4$ with $n \geq 1000$, 4-bit quantization produces a *negative* penalty ($\Delta = -0.004$ at $n = 1000$; -0.006 at $n = 5000$), meaning that at low rank and sufficient data, even 4-bit NF4 acts as a mild regularizer. Figure ?? shows this rank-dependent reversal clearly.

6 Discussion

Why quantization does not regularize at 1.1B. Three mechanisms counteract the theoretical $R(\Delta)$ benefit at small scale. **Capacity loss dominates:** quantizing a 1.1B base degrades intermediate representations on which LoRA adapters operate, and coherent instruction-following requires full-vocabulary fidelity that even NF4 partially destroys. **LoRA already regularizes sufficiently:** with $r \in \{4, 16, 64\}$ on a 1.1B-parameter model, the adapter introduces at most 22M free parameters against ≈ 51 K training tokens at $n = 100$ —an already favorable token-to-parameter ratio that leaves little room for further regularization. **Deterministic noise does not average out:** unlike stochastic regularizers (dropout, data augmentation), quantization error Δ is a fixed, input-independent perturbation applied identically at every step, introducing a constant gradient bias rather than a variance-reducing perturbation.

Why 8-bit regularizes at 7B. The picture changes fundamentally at 7B for two reasons. **Excess capacity:** a 7B FP16 model has roughly $6\times$ more parameters than TinyLlama. Fine-tuning on

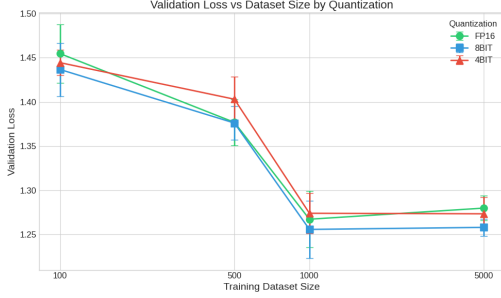


Figure 5: Mistral-7B: validation loss vs. training dataset size. 8-bit lies *below* FP16 at every data size, with the gap widening as n increases—the opposite of H2’s prediction.

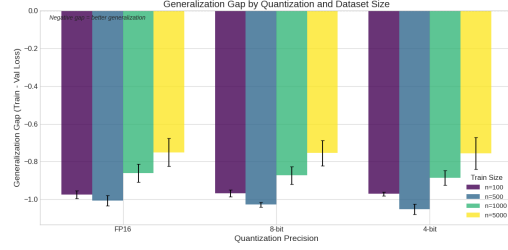


Figure 6: Mistral-7B: generalization gap by precision and training size. Unlike TinyLlama, the gaps differ more noticeably across precisions, with 4-bit exhibiting a distinctly larger-magnitude gap than 8-bit.

H3 (Mistral-7B): Quantization Effect by LoRA Rank

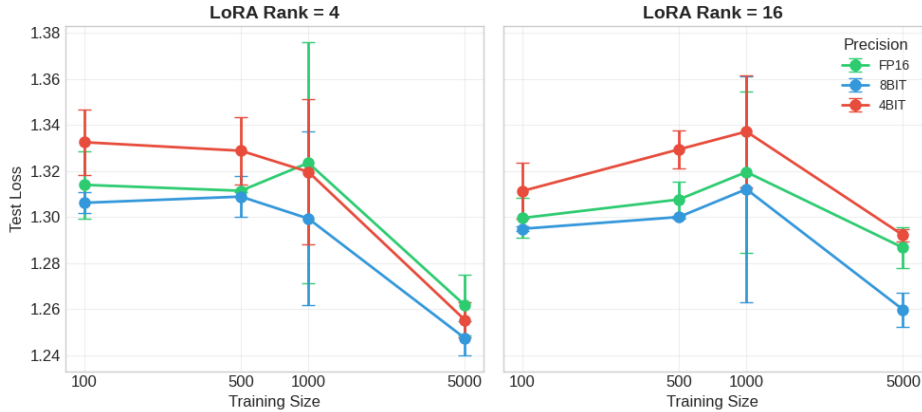


Figure 7: Mistral-7B H3: test NLL vs. training size stratified by LoRA rank ($r \in \{4, 16\}$). At $r = 4$ (left panel), 4-bit NF4 approaches and slightly beats FP16 at $n \geq 1000$. At $r = 16$ (right panel), 4-bit consistently hurts. H3 is clearly supported: lower rank substantially reduces and can reverse the quantization penalty.

100–5000 Alpaca examples leaves a vast amount of this capacity unutilized by the task; 8-bit’s mild capacity reduction restricts this excess without damaging core language representations. **Loss landscape:** larger models are known to have sharper loss landscapes. QGen’s theoretical argument—that quantization noise smooths the landscape toward flatter minima—has more to offer at 7B, where landscape sharpness is a real constraint. The growing-with- n pattern (H2 inverted) is a key signature of this mechanism: it is precisely when optimization has run long enough to reach sharp minima that flat-minima-inducing quantization becomes beneficial.

Why 4-bit still hurts at 7B (mostly). 4-bit NF4 represents a stronger compression than 8-bit, and at low data sizes ($n \leq 500$) the representational loss in the frozen base is large enough to outweigh the regularization benefit. Only when both data is sufficient ($n \geq 1000$) and rank is low ($r = 4$) does 4-bit cross into beneficial territory, suggesting a narrow operating regime where the adapter’s limited capacity forces it to use the noise-perturbed base representations constructively.

Practical implications. At $\geq 7B$ scale: 8-bit quantization should be the default choice—it saves $\approx 50\%$ VRAM and improves generalization by up to 0.021 NLL at $n = 5000$. 4-bit NF4 is viable when memory is the binding constraint, accepting a moderate performance cost (0.005–0.020 NLL at low n ; near-neutral at $n = 5000$ with $r = 4$). At sub-2B scale: choose quantization solely based on

memory budget; 8-bit incurs ≤ 0.003 NLL degradation; 4-bit consistently degrades by 0.012–0.019 NLL.

Limitations. The 7B scale-up uses only 2 seeds and 2 ranks, limiting statistical precision (the large std at $n = 1000$ FP16, ± 0.036 , reflects this). We test a single task (Alpaca instruction-tuning); structured-output or numeracy-heavy tasks may behave differently. The per-example H4 hypothesis (quantization harms numeracy/rare-token examples disproportionately) was not empirically evaluated. All experiments use post-training quantization; quantization-aware training (QAT) with learnable scales may behave differently.

7 Conclusion

We conducted a systematic empirical study of whether quantization noise in QLoRA-style parameter-efficient fine-tuning acts as implicit regularization in low-data instruction-tuning—and found that the answer depends critically on model scale. At TinyLlama-1.1B, 108 controlled experiments show no regularization benefit: FP16 consistently achieves the best or equivalent test loss, generalization gaps are identical across precisions, and the rank-interaction is at most marginal. At Mistral-7B-v0.1, 48 experiments reveal the opposite: 8-bit quantization outperforms FP16 at every data size, with the benefit growing as training data increases—a clear signature of the loss-landscape-smoothing mechanism proposed by ?. The rank-interaction hypothesis is also more strongly supported at 7B, with the 4-bit penalty being $2.2\times$ larger at $r = 16$ than at $r = 4$ and even turning negative at $r = 4$, $n \geq 1000$.

These findings reconcile a potential contradiction in the literature: the QGen mechanism is real but scale-gated. At small scales, LoRA’s low-rank constraint already provides sufficient implicit regularization and capacity loss from quantization dominates; at large scales, the model’s excess representational capacity makes 8-bit’s mild compression genuinely beneficial. Practitioners fine-tuning ≥ 7 B models should prefer 8-bit quantization as a default; those working with smaller models should choose precision based on memory constraints alone.

Acknowledgments and Disclosure of Funding

Experiments were conducted on Google Colab Pro using an NVIDIA L4 GPU (24 GB). We thank the developers of the Hugging Face Transformers, PEFT, bitsandbytes, and Datasets libraries for making this work accessible.

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Justification: The Stanford Alpaca dataset is publicly available via the Hugging Face Datasets hub (`tatsu-lab/alpaca`). Mistral-7B-v0.1 is fully open under the Apache 2.0 license (`mistralai/Mistral-7B-v0.1`). The full experimental notebook (`analysis.ipynb`) with all training, evaluation, and plotting code is included in the project repository.

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6. Experimental Setting/Details

Question: Does the paper specify all the training details (e.g., data splits, done hyperparameter search, best model selection)?

Answer: [\[Yes\]](#)

Justification: Section 4 provides complete training details for both TinyLlama-1.1B and Mistral-7B-v0.1. No hyperparameter search was performed; all hyperparameters are fixed following standard QLoRA recipes. We report results for all configurations, not just selected best runs.

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7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined and does the paper report confidence intervals or statistical significance?

Answer: [\[Yes\]](#)

Justification: All reported means include ± 1 standard deviation across seeds (and LoRA ranks where noted). Three seeds per configuration at 1.1B and two seeds at 7B allow assessment of run-to-run variance. Formal hypothesis tests were not conducted; we rely on the consistency of results across all 156 total runs.

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Question: For each experiment, does the paper provide sufficient information on the computational resources (type of compute, memory, time) for reproducing experiments?

Answer: [\[Yes\]](#)

Justification: Section 4 specifies the hardware (NVIDIA L4 GPU, 24 GB, Google Colab Pro) and total number of runs (108 at 1.1B, 48 at 7B). Per-run estimates can be derived from the reported dataset sizes and model sizes.

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models?

Answer: [\[NA\]](#)

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